

Design and Implementation of a Vision-Guided Robotic System for Automated Vegetable Sorting

Fathhur Rahaman Sams, Sanjana Kazi Supti, Md Junaeid Ali, Mahfuj Alam Imon
Department of Computer Science & Engineering, East West University

Motivation

- The paper introduces a vision-enabled robot-based system designed to perform automation for vegetables sorting by means of freshness detection.
- A specialized image dataset containing 4,640 pictures of different types of vegetables (eight in total).
- Both in their fresh state and with damage present, has been created.
- Modern algorithms based on YOLO technology have been examined.
- YOLOv12s became a baseline for this task.
- Self-supervised learning strategies (DINOv2-like and BYOL) have been employed to improve detection quality and minimize reliance on labeling.
- The enhanced model was combined with a 6-DOF robotic arm for performing sorting activities.

Methodology

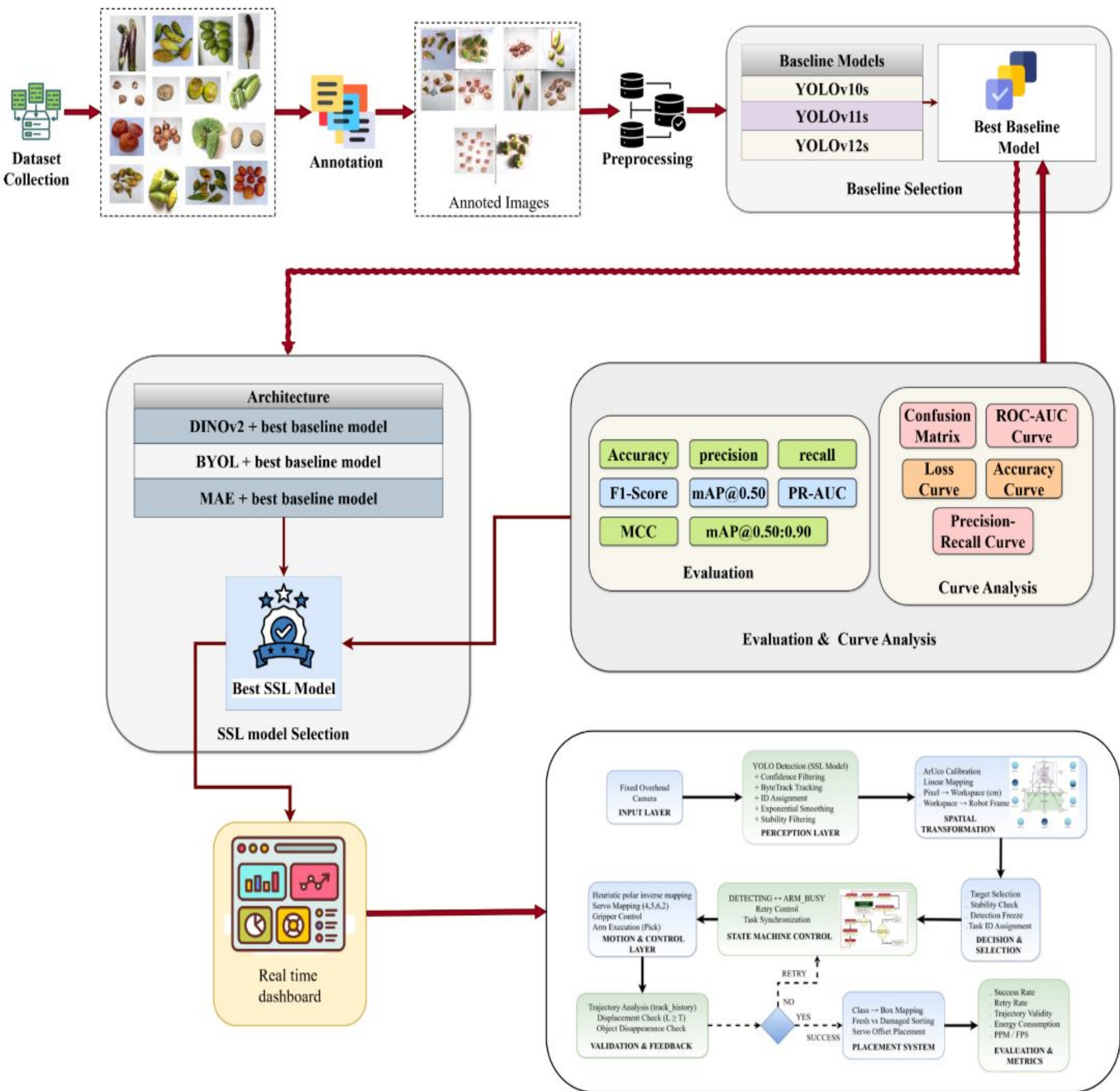


Fig 1: Workflow Diagram

Dataset and SSL Architecture

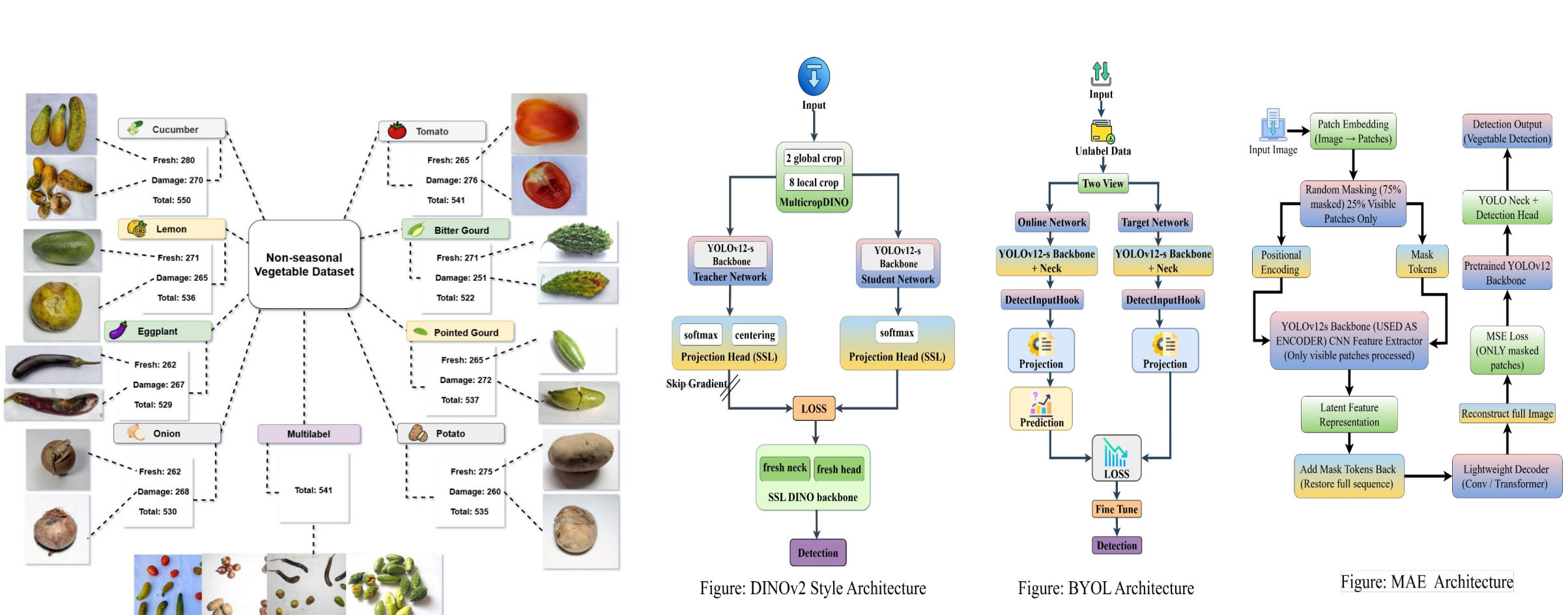


Fig 2: Dataset Workflow

Fig 3: DINOv2, BYOL and MAE Architecture

Vision-Guided Robotic Automation Pipeline

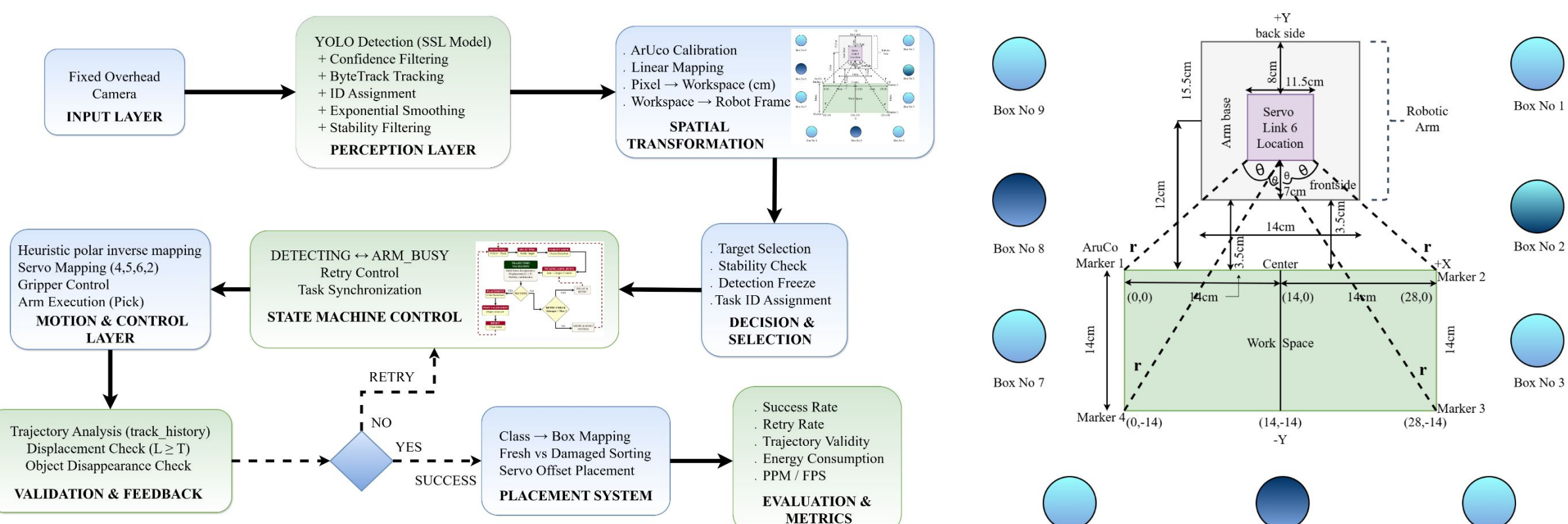


Fig 4: System Architecture Overview

Fig 5: System Workspace

Experiments & Result Analysis

- YOLOv12s outperforms previous models in detection accuracy.
- Self-supervised learning further boosts performance.
- DINOv2 achieves the best results (>99% precision, mAP@0.5).
- Model remains robust under lighting variations and occlusions.
- Real-time tests confirm accurate classification and sorting.

Table 1: SSL Architecture Performance With Best Baseline Backbone

Architecture	Model	Precision	Recall	F1-Score	mAP50	mAP50-95
BYOL	YOLOv12s	97.6%	96.8%	97.2%	98.4%	83.1%
DINOv2-style	YOLOv12s	99.1%	98.5%	98.8%	99.6%	88.3%
MAE	YOLOv12s	98.32%	99.22%	98.76%	98.85%	84.69%

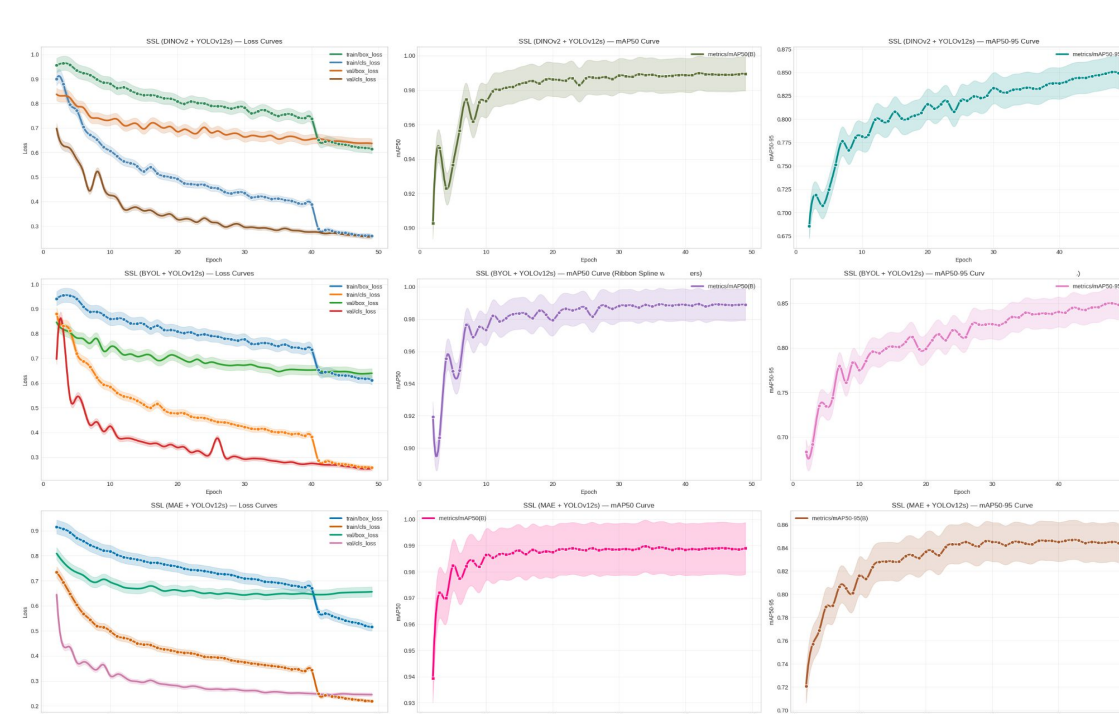


Fig 6: Loss and mAP Curve of SSL Architectures

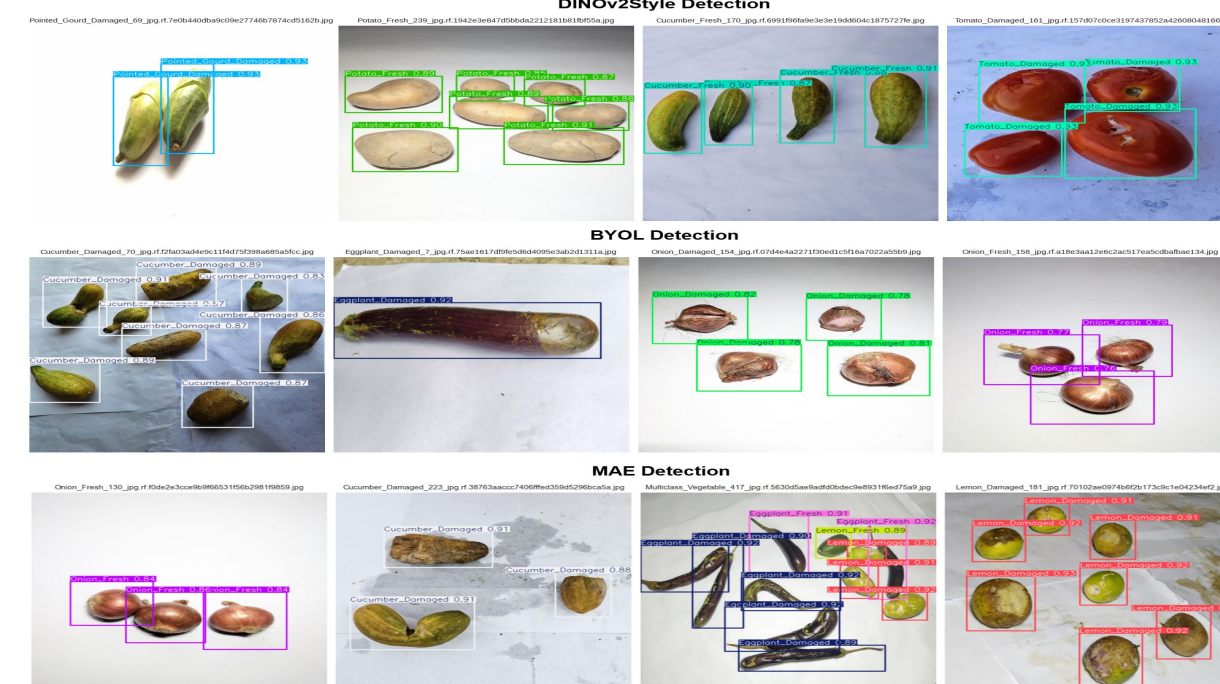


Fig 7: Vegetable Detection Visualized by SSL Architecture

Table 2: Class-wise pick-and-place performance over a 5-minute evaluation period

Class	Success	Fail	Total	Success Rate (%)
Bitter_Gourd_Fresh	4	1	5	80.0
Cucumber_Fresh	5	0	5	100.0
Eggplant_Fresh	4	0	4	100.0
Lemon_Fresh	5	0	5	100.0
Onion_Fresh	4	1	5	80.0
Pointed_Gourd_Fresh	5	0	5	100.0
Potato_Fresh	5	0	5	100.0
Tomato_Fresh	5	2	7	71.4
Damaged	6	1	7	85.7
Total	43	5	48	89.6

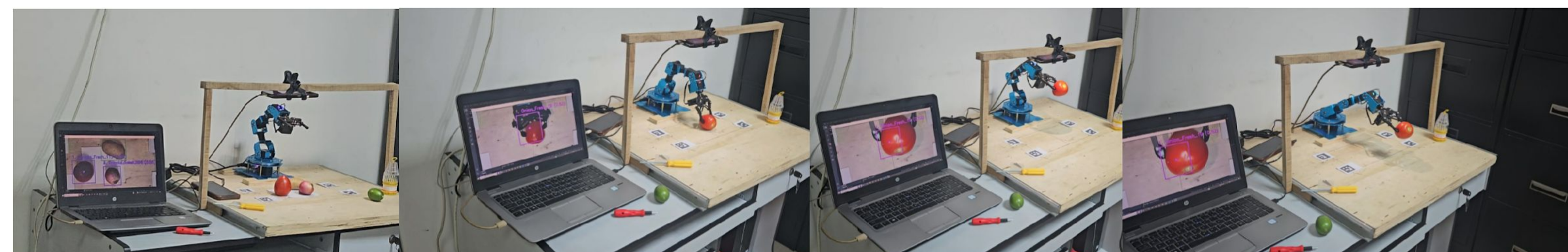


Fig 6: Sequential Operational Workflow

Table 3: Core system performance metrics computed from runtime execution

Metric	Value
Total Pick Cycles	48
Successful Picks	43
Success Rate (%)	89.6
Average FPS	29.0
Tasks per Minute (PPM)	9
Tasks per 5 Minutes	48
Average Pick Duration (s)	3.10
Average Attempts per Task	1.18
Retry Rate	1.18
Max Retry Failure Count	5

Table 4: Detection and tracking performance metrics

Metric	Value
Total Detections	320
Stable Detections	280
Detection Stability Ratio (%)	87.5
Track Loss Count	20
Track Loss Rate (%)	6.25
Disappearance-Based Success Rate (%)	89.6

Conclusions

This research provides a fully integrated robot vision system for vegetable sorting automation. The use of self-supervision and robust detection models ensures that the proposed method is accurate while minimizing the labeling process. This combination of perception, explainability, and robot control highlights an innovative way forward for smart farming applications. Going forward, we will concentrate on making our solution more adaptable and efficient for real-world industrial use cases.

References

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